

Department of Computer Engineering
Faculty of Engineering

Kasetsart University

HW#3: SQL Database Implementation

Choose your **own domain** (e.g., Hospital, School, Restaurant, etc.) to design and implement your database system (or a library management system).

Foreign Key Declaration Example:

Before you start, here's how to declare foreign keys in SQL:

```
-- Example: Library Management System Tables
CREATE TABLE Author (
    authorID INT PRIMARY KEY,
    authorName VARCHAR(255),
    birthYear INT
);

CREATE TABLE Book (
    bookID INT PRIMARY KEY,
    title VARCHAR(255),
    authorID INT,
    category VARCHAR(100),
    publishYear INT,
    -- Declare foreign key reference
    FOREIGN KEY (authorID) REFERENCES Author(authorID)
);

CREATE TABLE Member (
    memberID INT PRIMARY KEY,
    memberName VARCHAR(255),
    email VARCHAR(255),
    joinDate DATE
);

CREATE TABLE Borrowing (
    borrowID INT PRIMARY KEY,
    memberID INT,
    bookID INT,
    borrowDate DATE,
    returnDate DATE,
    -- Declare multiple foreign keys
    FOREIGN KEY (memberID) REFERENCES Member(memberID),
    FOREIGN KEY (bookID) REFERENCES Book(bookID)
);
```

Key Points:

- Foreign keys create relationships between tables
- FOREIGN KEY (column_name) REFERENCES referenced_table
- The referenced column must be a primary key in the referenced table
- Foreign keys ensure referential integrity

Requirements:

1.1) Database Design and Creation

- Use DDL-SQL to create your database
- Design at least 4 related tables
- Declare appropriate data types for all columns
- Define primary keys and foreign keys
- Include at least one table with a composite primary key

1.2) Data Population

- Use DML-SQL INSERT statements to populate your database
- Insert at least 5-10 instances for each table
- Ensure data consistency and referential integrity
- Include diverse data that will make your queries interesting

1.3) Business Intelligence Queries

Write 10 business questions and their corresponding SQL queries. Your queries must demonstrate the following SQL concepts covered in class:

Required Query Types:

1. Basic Selection with WHERE clause (1 query)
2. Cartesian Product Operations (2 queries)
3. Aggregation Functions (min, max, sum, avg, count) (2 queries)
4. GROUP BY with HAVING clause (1 query)
5. Set Operations (UNION, INTERSECT, or EXCEPT) (1 query)
6. Table Variables - using table aliases (1 query)
7. Theta-join Operations - joins with <, <=, >, >=, <> conditions (1 query)
8. ORDER BY with multiple columns (1 query)
9. LIKE operator with wildcards (1 query)
10. Complex query combining multiple concepts (1 query)

Example Business Questions for Library System:

- Which books have been borrowed more than 5 times?
- What is the average number of books borrowed per member by city?
- Which authors have books in multiple categories?
- Find members who have borrowed both 'Fiction' and 'Science' books
- What are the top 3 most popular book categories by borrowing frequency?
- Find books that are more expensive than the average book price (theta-join)
- Show borrowing records ordered by member name and then by borrow date

Submit the following:

- **HW#3_StudentID.sql**
- **HW#3_StudentID.pdf (Document with queries and results)**

Note: Plagiarism will result in zero points. Each student must create their own unique database schema and business questions.